ISSUED FOR APPROVAL

4_AHU-03 (A)

ELECTRICAL DATA:		
Rated Operating voltage	V AC	400
Rated frequency	Hz	60
Rated current	A	100A
System Configuration		TNS, 3 PHASE, 4 WIRES
Rated control voltage	V AC	230
Rated control voltage	V DC	N/A
Rated insulation voltage	V	1000
Rated Impulse Withstand Voltage	KV	8

BUSBAR DETAILS:		
Main busbar cross section	mm	1x2x20x10
Neutral Busbar	mm	1x2x20x10
Ground Busbar	mm	1x2x20x5
Short Time Current (kA) for 1sec	Icw	25
Busbar treatment		Tin-plated Copper
Busbar Insulation		No
Color code		Red, Yellow, Blue, Black, Green/Yellow
Version of external connection		L1,L2,L3,N,PE

ENCLOSURE DETAILS:		
Type of Switchgear		Sivacon S8
Installation		Free Stand
Internal separation		Form 4b
Degree of protection	IP	43
Canopy		Yes
Ventilation		Yes
Gland plate		Yes
Color of enclosure		RAL 9002
Color of canopy		RAL 9002

CONNECTION DETAILS:		
Cable connection for supply cubicle(s)		Top
Cable connection for outgoing cubicle(s)		Top
Connection in plant		Front

SERVICE LOCATION:		
Installation Area		Indoor
Ambient temperature	°C	40
Altitude	m	<2000

WIRING:		
Wires type		PVC 450-750 V, Cu
Control circuit cross section	mm²	1.5
CT circuit cross section	mm²	2.5
Auxiliary power circuit cross section	mm²	1.5
AC circuit color		RED, BLACK
DC circuit color		N/A
CT circuit color		RED, YELLOW, BLUE
Auxiliary power circuit color		RED, BLACK
Labeling		Cable Numbers

IDENTIFICATION LABEL:		
Material		Gravoply
Color		White Text in Black Background
Language		English
Fixation		Adhesive Glue

REVISION

REV	DESCRIPTION	PREP	CHKD	APPD	DATE
R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

PROJECT
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DRAWING TITLE
DATA SHEET

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 1 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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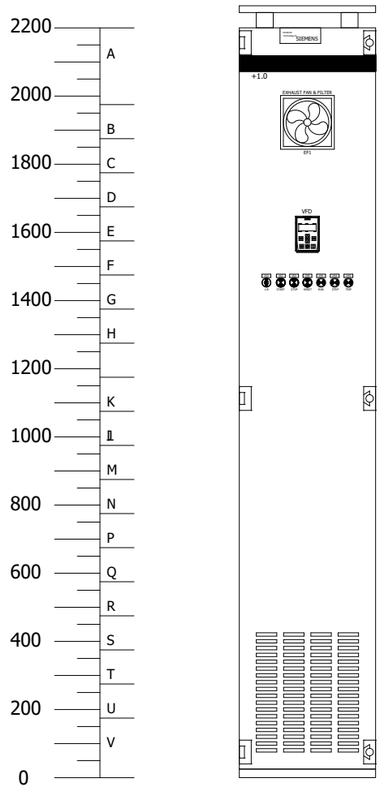

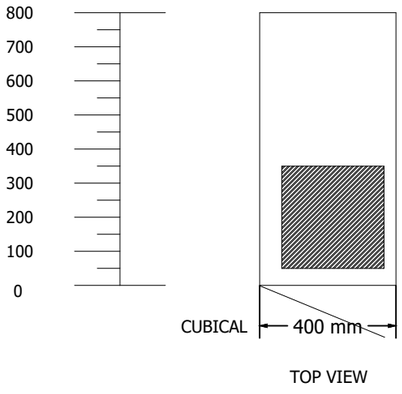

TOP VIEW

REVISION

REV	DESCRIPTION	PREP	CHKD	APPD	DATE
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DRAWING TITLE

GENERAL ARRANGEMENT

MANUFACTURER

QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 2 of 18

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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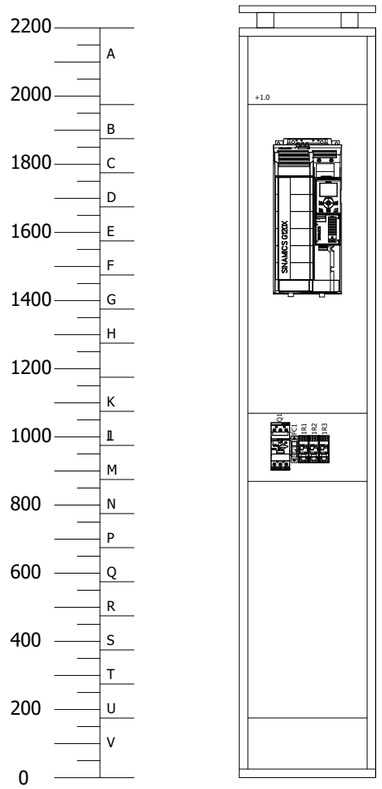

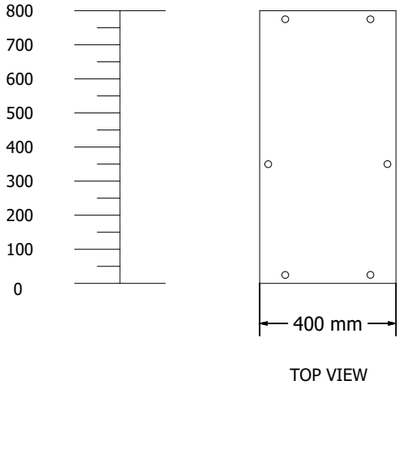

TOP VIEW

REVISION					
REV	DESCRIPTION	PREP	CHKD	APPD	DATE
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DRAWING TITLE
GENERAL ARRANGEMENT

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 3 of 18

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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REVISION					
REV	DESCRIPTION	PREP	CHKD	APPD	DATE
R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

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DRAWING TITLE

CLIENT SINGLE LINE DIAGRAM

MANUFACTURER

QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 4 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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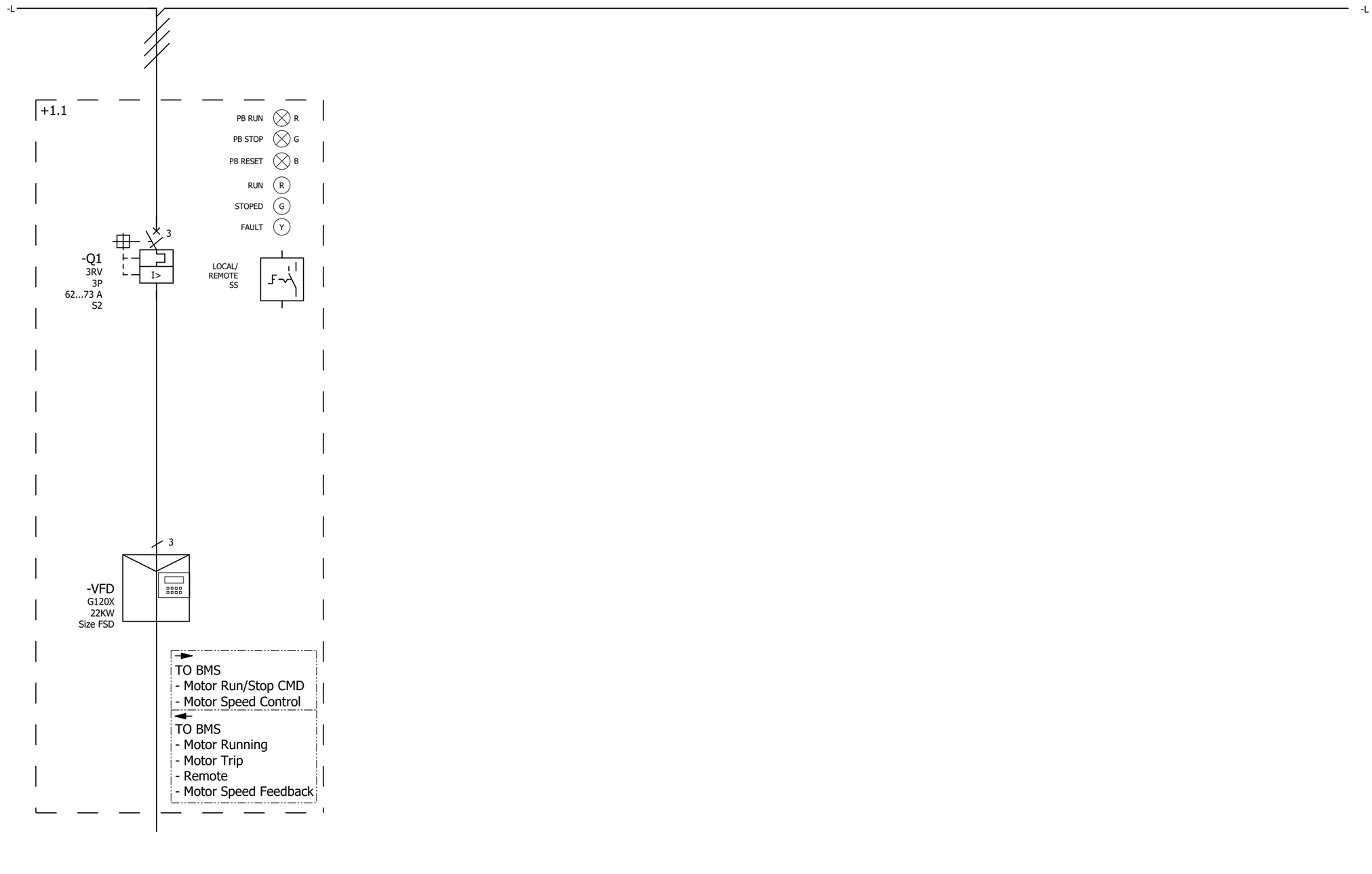
100A, 400/230V; 3Ph. + N & E; 4W; 60HZ,17KA/1Sec.



AMP. RATING	-
LOAD	22KW
SERVICE	-
CABLE SIZE	-

REVISION

REV	DESCRIPTION	PREP	CHKD	APPD	DATE
R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

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DRAWING TITLE
SINGLE LINE DIAGRAM

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 5 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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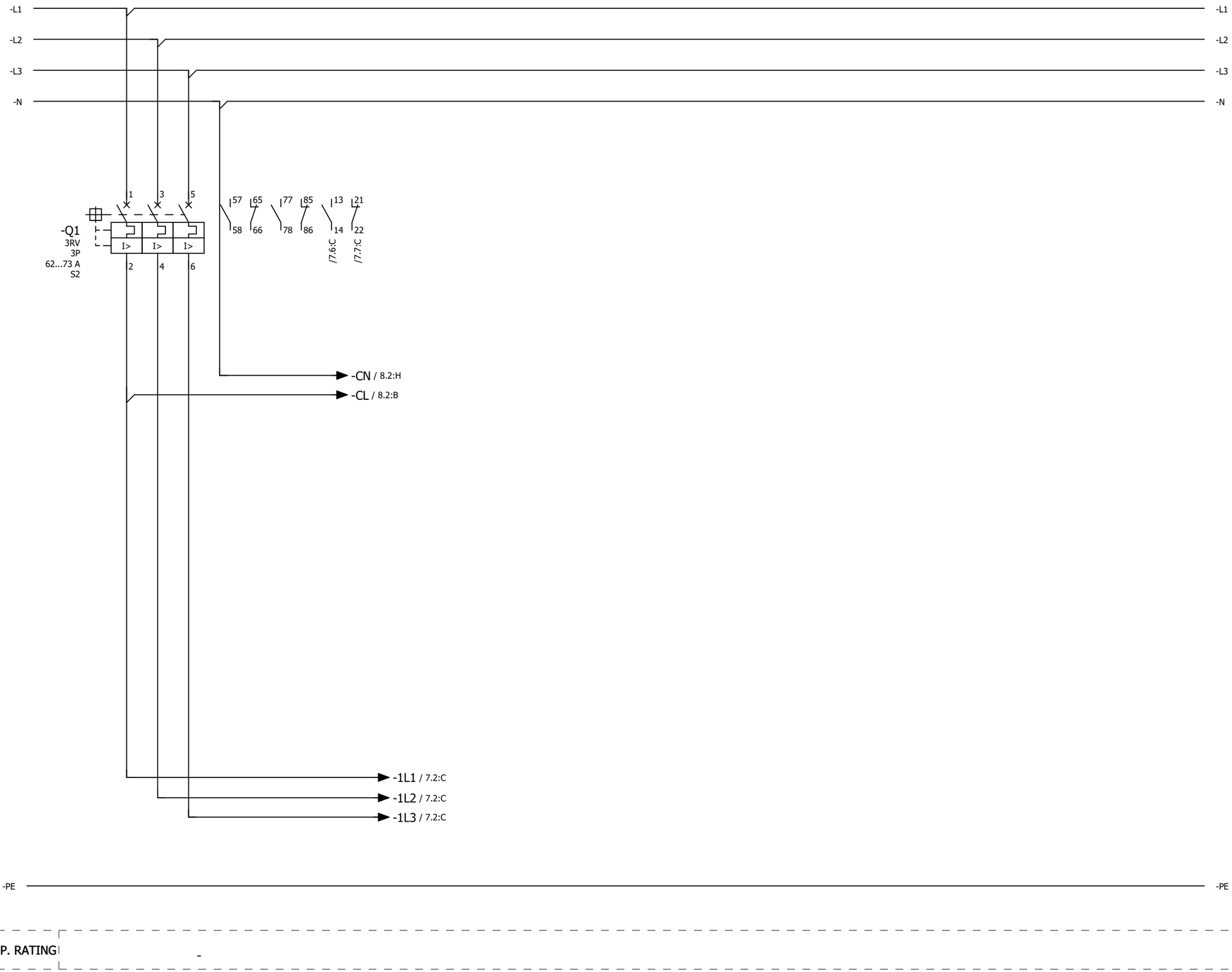
100A, 400/230V; 3Ph. + N & E; 4W; 60HZ,17KA/1Sec.



AMP. RATING	-
LOAD	22KW
SERVICE	-
CABLE SIZE	-

REVISION

REV	DESCRIPTION	PREP	CHKD	APPD	DATE
R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

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DRAWING TITLE
SCHEMATIC DIAGRAM

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 6 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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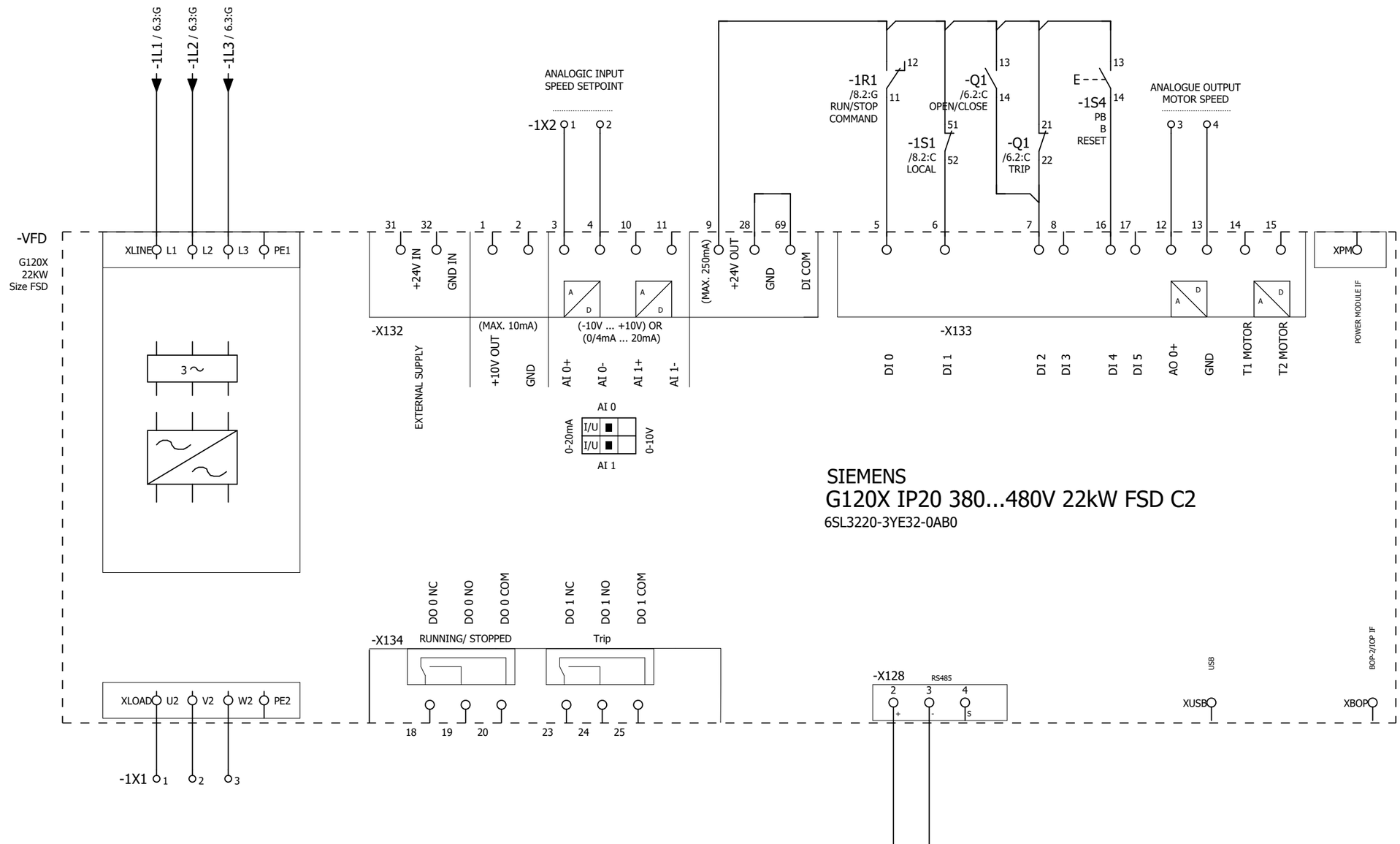

REVISION					
REV	DESCRIPTION	PREP	CHKD	APPD	DATE
R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

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DRAWING TITLE
SCHEMATIC DIAGRAM

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 7 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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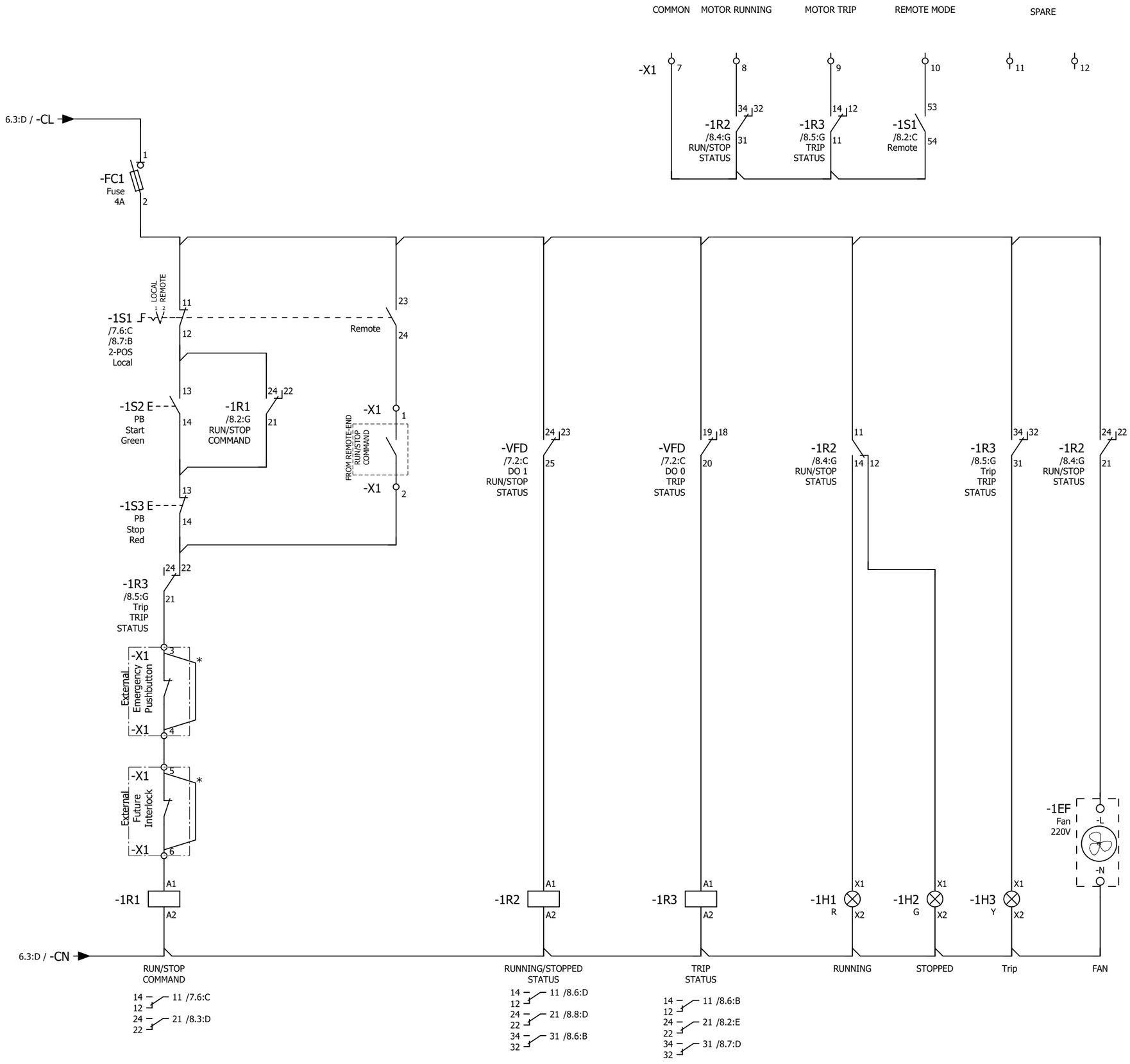

REVISION

REV	DESCRIPTION	PREP	CHKD	APPD	DATE
R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

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DRAWING TITLE
SCHEMATIC DIAGRAM

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 8 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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Terminal diagram

Strip: =4_AHU-03 (A)+-X1

External targets					Internal targets			
Target designation	Wiring	Jumper	SD	Jumper	Wiring	Target designation	Placement	Function text
			3			-1R3:21	/8.2:E	External Future Interlock
			4				/8.2:F	=
			5					=
-1R1:A1			6				/8.2:G	=
-1R2:31			7					TO PLC - COMMON
-1R2:34			8					MOTOR RUNNING
-1R3:14			9					MOTOR TRIP
-1S1:53			10					REMOTE MODE
			11					SPARE
			12					=
			1			-1S1:24	/8.4:C	FROM REMOTE-END RUN/STOP COMMAND
-1R3:22			2				/8.4:D	=

Strip: =4_AHU-03 (A)+-1X1

External targets					Internal targets			
Target designation	Wiring	Jumper	SD	Jumper	Wiring	Target designation	Placement	Function text
			1			-VFD:XLOAD:U2	/7.2:F	POWER TERMINAL
			2			-VFD:XLOAD:V2	/7.2:F	=
			3			-VFD:XLOAD:W2	/7.2:F	=

Strip: =4_AHU-03 (A)+-1X2

External targets					Internal targets			
Target designation	Wiring	Jumper	SD	Jumper	Wiring	Target designation	Placement	Function text
			1					SPARE
			2					ANALOGIC INPUT SPEED SETPOINT 4-20mA
			3					ANALOGIC OUTPUT SPEED FEEDBACK 4-20mA
			4					=

Strip: =4_AHU-03 (A)+-XPM

External targets					Internal targets			
Target designation	Wiring	Jumper	SD	Jumper	Wiring	Target designation	Placement	Function text
			1			-VFD:XLOAD:2	/7.6:F	MODBUS RTU CONNECTION (-)
			2			-VFD:XLOAD:3	/7.6:F	MODBUS RTU CONNECTION (+)

REVISION

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R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

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DRAWING TITLE
TERMINAL LINE-UP DIAGRAM

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.
PROJECT REFERENCE:
REC. 00
SCALE : NTS
LOCATION: +
SIZE: A4
SHEET NO: 9 of
QUOTATION REF.
FILE NAME: 4_AHU-03 (A)

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Device tag list

-1EF	/8.8:G
NTL-SF150-230B (SFF004)	1

-FC1	/8.2:B
3NW6004-1	1
3NW7013	1

-1H1	/8.6:G
3SB6216-6AA20-1AA0	1

-1H2	/8.7:G
3SB6216-6AA40-1AA0	1

-1H3	/8.7:G
3SB6216-6AA30-1AA0	1

-Q	/7
6SL3220-3YE30-0AB0	1

-Q1	/6.2:C
3RV2031-4KA10	1
3RV2901-1E	1
3RV2921-1M	1

-1R1	/8.2:G
LZX:PT570730	1
LZS:PT78740	1

-1R2	/8.4:G
LZX:PT570730	1
LZS:PT78740	1

-1R3	/8.5:G
LZX:PT570730	1
LZS:PT78740	1

-1S1	/8.2:C
3SB6160-4AA11-1BA0	1
3SB6400-1AA10-1CA0	2
3SB6400-1AA10-1BA0	1

-1S2	/8.2:D
3SB6210-0AB40-1FA0	1

-1S3	/8.2:E
3SB6210-0AB20-1FA0	1

-1S4	/7.7:C
3SB6210-0AB10-1FA0	1

-VFD	
6SL3220-3YE32-0AB0	1
6SL3256-0AP00-0JA0	1

REVISION

REV	DESCRIPTION	PREP	CHKD	APPD	DATE
R00	FOR APPROVAL	STM	SAK	SAK	20/03/25

PROJECT
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DRAWING TITLE
DEVICE TAG LIST

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 10 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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Bill of Material

Order number	Qty.	Description	Make
NTL-SF150-230B (SFF004)	1	Filter fan, Extract: W: 125 mm, H: 125 mm, RAL 7035, IP54, Airflow: 63 m3/h, Ue AC: 230 V, 150×150×80.1	NATURAL
3NW6004-1	1	CYLINDRICAL FUSE GG ACC. TO FRENCH STANDARD (NFC) WITHOUT INDICATOR SIZE 10X38MM, 500V 4A	SIEMENS
3NW7013	1	SENTRON, CYLINDRICAL FUSE HOLDER, 10X38 MM, 1-POLE, IN: 32 A, UN AC: 690 V	SIEMENS
3SB6216-6AA20-1AA0	1	COMPACT PILOT LIGHT 22 MM PLASTIC ROUND INTEGRATED LED 220V AC RED	SIEMENS
3SB6216-6AA40-1AA0	1	COMPACT PILOT LIGHT 22 MM PLASTIC ROUND INTEGRATED, LED, 220V AC, GREEN	SIEMENS
3SB6216-6AA30-1AA0	1	COMPACT PILOT LIGHT 22 MM PLASTIC ROUND INTEGRATED LED 220V AC YELLOW	SIEMENS
6SL3220-3YE30-0AB0	1	SINAMICS G120X Rated power: 18.5 kW At 110% 60s, 100% 240 s Radio interference suppression filter for category C2 380-480 V 3 AC +10/-20% 47-63 Hz Ambient temperature -20 to +45 °C Size: FSD Degree of protection IP20 / UL open type	SIEMENS
3RV2031-4KA10	1	Circuit breaker size S2 for motor protection, CLASS 10 A-release 62...73 A N-release 949 A screw terminal Standard switching capacity	SIEMENS
3RV2901-1E	1	TRANSVERSE AUX. SWITCH, 1NO+1NC, SCREW CONNECTION, FOR CIRCUIT-BREAKERS 3RV2	SIEMENS
3RV2921-1M	1	SIGNALING SWITCH, FOR CIRCUIT-BREAKERS 3RV2. W. SCREW CONNECTION, ,	SIEMENS
LZX:PT570730	3	Plug-in relay, 4 changeover contacts 230 V AC, 6 A, Width 22.5 mm for LZS sockets	SIE
LZS:PT78740	3	BASE FOR RELAY WITH 4 CHANGEOVER CONTACTS	SIEMENS
3SB6160-4AA11-1BA0	1	Complete unit combination 22 mm metal shiny round key selector switch 2 switch positions KL-KR with 2 keys 90DEG, V position latching removal pos. left and right with holder screw terminal, 1 NO	SIEMENS
3SB6400-1AA10-1CA0	2	CONTACT BLOCK ACTUATOR COMPONENT WITH 1 CONTACT ELEMENT SCREW TERMINAL, 1 NC	SIEMENS
3SB6400-1AA10-1BA0	1	CONTACT BLOCK ACTUATOR COMPONENT WITH 1 CONTACT ELEMENT SCREW TERMINAL, 1 NO	SIEMENS
3SB6210-0AB40-1FA0	1	COMPACT PUSHBUTTON 22 MM PLASTIC BLACK ROUND SCREW TERMINAL, 1 NO + 1 NC GREEN	SIEMENS
3SB6210-0AB20-1FA0	1	COMPACT PUSHBUTTON 22 MM PLASTIC BLACK ROUND SCREW TERMINAL, 1 NO + 1 NC RED	SIEMENS
3SB6210-0AB10-1FA0	1	Compact pushbutton 22 mm plastic black round screw terminal, 1 NO + 1 NC black	SIEMENS
6SL3220-3YE32-0AB0	1	SINAMICS G120X Rated power: 22 kW At 110% 60s, 100% 240 s Radio interference suppression filter for category C2 380-480 V 3 AC +10/-20% 47-63 Hz Ambient temperature -20 to +45 °C Size: FSD Degree of protection IP20 / UL open type	SIEMENS
6SL3256-0AP00-0JA0	1	SINAMICS G120 DOOR MOUNTING KIT IP54/UL TYP12 FOR INTELLIGENT OPERATOR PANEL AND BASIC OPERATOR PANEL BOP-2 CONTAINS SEAL, MOUNTING MATERIAL AND CONNECTION CABLE (5M)	SIEMENS
CTS2.5UN	28	2.5 SQ.MM, 5 MM THICK FEED THROUGH TERMINAL BLOCK	CONNECTWELL
CTS6UN	3	6MM sq.mm, Feed Through Terminal Block	CONNECTWELL

REVISION

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CONTRACTOR
SAFARI سفاري

DRAWING TITLE
SUMMARIZED PARTS LIST

MANUFACTURER
QETAF ELECTRICS
The Highest Quality

DRAWING NO.

PROJECT REFERENCE:

REC. 00

SCALE : NTS

LOCATION: +

SIZE: A4

SHEET NO: 11 of

QUOTATION REF.

FILE NAME: 4_AHU-03 (A)

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